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Tomorrow's tech

At Moore's end: After five decades of driving innovation

After fifty years of following the Moore's law, computing is poised to take the quantum leap.

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Here's a thought: if the number of people in your house were to double every year, you will end up with 512 people in ten years, provided you started off living alone. Here's another one: if you were spending a thousand bucks for yourself in year one and thousand bucks for 512 people in year ten, that would mean less than two bucks expense per person. Lots of savings, maybe, but your house will be somewhat overcrowded, possibly leading to fights and struggles. The probable future of Moore's Law is similar.

Back on April 19, 1965, Gordon Moore, a researcher at Fairchild Semiconductor wrote a paper predicting the pace of growth of technology. He said, transistors on a single chip will double annually, while cost of the chip will remain constant. This was fifty years ago, when technology was nascent. Moore's prediction was based on observations drawn from components used per integrated circuit (IC), and respective minimum cost-per-component, between 1959 and 1964. In 1975, he predicted that each chip will contain up to 65,000 components, dou-

bling annually within the time span. From 1975, technology more or less followed Moore's observation. His predictions became a reference point for researchers and developers to push processing power and mobility solutions beyond existing threshold.

But there comes time when everything reaches its peak. At the 2011 International Supercomputing Conference it was conceded that the Moore's Law might not be sufficient to drive processing innovation rapidly forward any longer. We are at a time where simply adding more transistors to a chipset and increasing the number of cores in a processor does not ensure viable progress in technology. Most of our tasks run neatly in both PCs and mobile devices sporting four-core and eight-core processors. The 12-core and 16-core processors of the future may be theoretically feasible, but do we really need them?

There is an issue with simply adding transistors to a chip. Once the number of layers of components in a chipset hits a certain threshold, Heisenberg's Uncertainty Principle will kick in. This will lead to a breakdown of subatomic particle paths, causing electron leaks and shorting circuits. Massively powerful processors that can melt down any time? Certainly doesn't sound like the future.

The Impetus and the Drive

The Intel 4004, Intel's first microprocessor, featured 2,250 transistors back

in 1971. Three decades later, the Intel Pentium 4 HT processor sported 125 million transistors – more than 50,000 times more than the Intel 4004.

In 1974, Robert Dennard at IBM presented transistor scaling equations to reduce operating factors along with dimensions, thereby shrinking transistor sizes. Another major milestone of Moore's form factor-shrinking prediction is Apple's introduction into the technology world. The co-founder of Apple, Stephen Wozniak, designed a device in 1975 that would go on to become the Apple I Personal Computer, which was followed by the Apple II. These two personal computers raised microprocessor and chipset demands, calling for redesigning of circuit boards and shrinkage of component sizes. In 1979, Intel introduced the 8088 processor featuring 29,000 transistor units.

The Intel 8088 notably featured in the now-legendary IBM PC, back in 1981. A year later, Intel revealed the 80826 processor (with 1,34,000 transistors), keeping up with Moore's prediction of the rate of advancement of technology. Components were not only getting smaller by dimensions, they were becoming more efficient in terms of energy, expenses, assembling and optimum power supply. Towards the end of the '80s, Intel unveiled the Intel 486 processor, with 1.2 million transistors – by far the most advanced microprocessor at that time.

The '90s were the era of Intel's Pentium, alongside the popularisation of port-